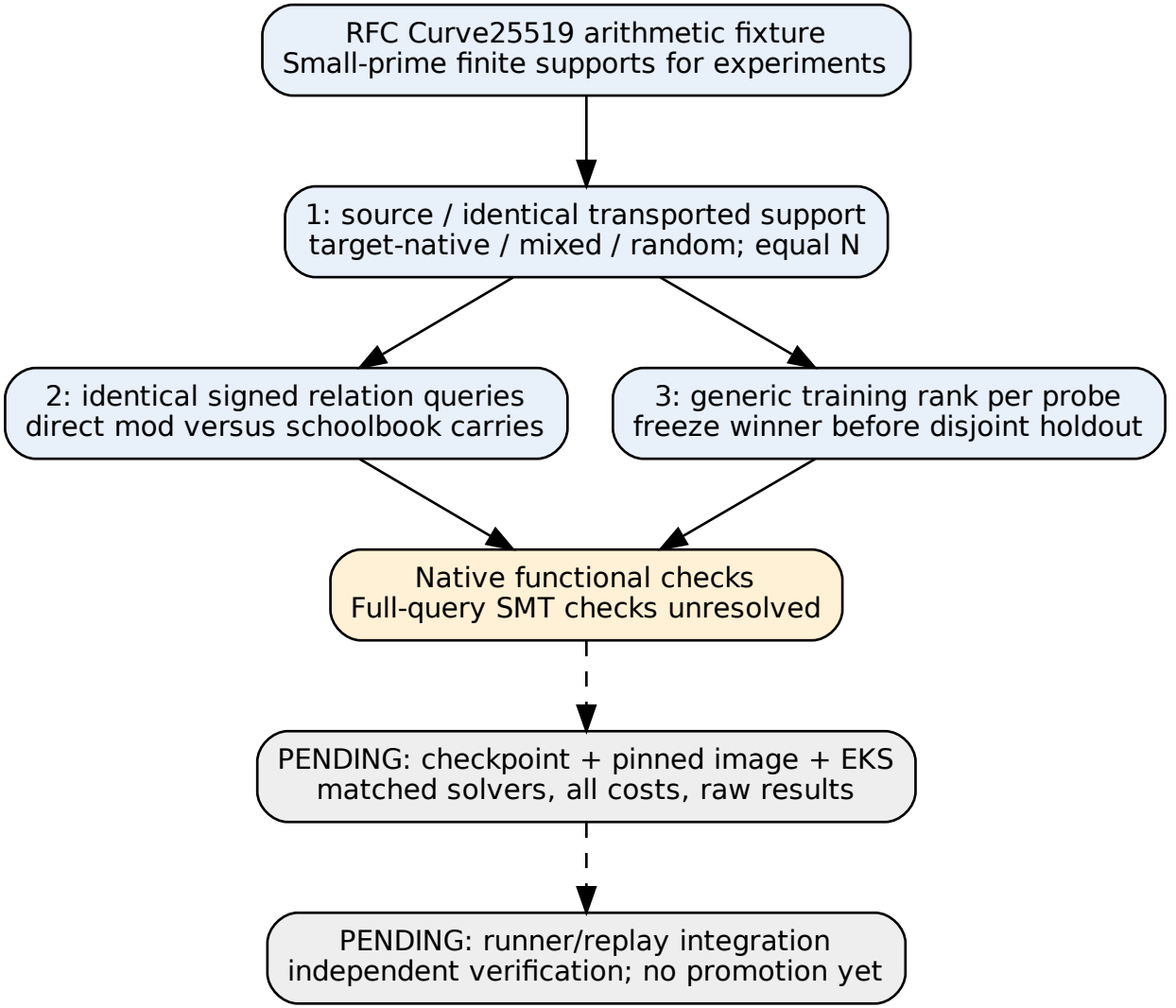Atlas / carry / rank: development and pending comparisons